# Testing Linear Hypotheses

**Lecture 4 (R code)**

## Table of contents


Reading . . . 2

**Hypothesis Tests** **2**

Simple hypotheses . . . 2
Linear combinations . . . 3
Multiple linear combinations . . . 3
Single linear combinations . . . 3
Example: PM2.5 . . . 4
Null hypotheses . . . 4
Alternative hypothesis . . . 5
Test statistic . . . 5
Variance . . . 6
Distribution . . . 7
Example: PM2.5 . . . 7
Multiple linear hypotheses . . . 7
Example: PM2.5 . . . 8
Test . . . 9
Test statistic . . . 9
Test statistic . . . 10
Distribution . . . 10
Rejection rule . . . 10
Alternative form . . . 11
Example: PM2.5 . . . 11
Revisiting single linear restriction . . . 12
Example: PM2.5 . . . 12
Regression significance . . . 13
Stata output . . . 13

**Structural Change** **14**
Dummy variables: recap . . . 14
Dummy variable interactions . . . 14
Example . . . 15
$F$-test . . . 17
Test: . . . 18
Null hypothesis . . . 18
Chow test . . . 19
Distribution . . . 19
Approach 2 . . . 19
Interactions . . . 20
Test . . . 20
$F$-stat . . . 21
Equivalence . . . 21
Demonstration . . . 21

**References** **23**
Bibliography . . . 23

### Reading

This lecture covers material from Wooldridge (2025):

- Chapter 4.4 - "Testing Hypotheses about a Single Linear Combination of the Parameters"
- Chapter 4.5 - "Testing Multiple Linear Restrictions"
- Chapter 7.4 - "Interactions Involving Dummy Variables"

Note: I have edited the syllabus slightly to move the topic of heteroskedasticity to Week 5 with asymptotics.

## Hypothesis Tests

### Simple hypotheses

So far, we have considered hypotheses that involve a single estimator:

1. $H_0 : \beta_j = \kappa_0$
2. $H_0 : \beta_j \geq \kappa_0$
3. $H_0 : \beta_j \leq \kappa_0$

## Linear combinations

Today, we consider hypotheses of the form:

$$H_0 : a_0\beta_0 + a_1\beta_1 + \cdots + a_k\beta_k = \kappa_0$$

for constants $\{a_j\}_{j=0,1,\ldots,k}$. This is a **linear combination of parameters**.

For example, $H_0 : \beta_1 = 2\beta_2 - 0.5$ can be written as,

$$H_0 : \underbrace{0}_{a_0}\,\beta_0 + \underbrace{1}_{a_1}\,\beta_1 + \underbrace{(-2)}_{a_2}\,\beta_2 = \underbrace{-0.5}_{\kappa_0}$$

## Multiple linear combinations

Similarly, we could have $q$ of these statements,

$$\begin{aligned} H_0 : a_0^1\beta_0 + a_1^1\beta_1 + \cdots + a_k^1\beta_k &= \kappa_0^1 \\ a_0^2\beta_0 + a_1^2\beta_1 + \cdots + a_k^2\beta_k &= \kappa_0^2 \\ &\vdots \\ a_0^q\beta_0 + a_1^q\beta_1 + \cdots + a_k^q\beta_k &= \kappa_0^q \end{aligned}$$

## Single linear combinations

Consider the following example related to environmental and health economics

$$\ln(PM2.5_\ell) = \beta_0 + \beta_1 EV_\ell + \beta_2 hybrid_\ell + \mathbf{x}_\ell\gamma + u_\ell$$

where,

- $PM2.5$ is micrograms of particulate matter (with diameter $\leq 2.5$ micrometres) per cubic metre in location $\ell$.
  - standard measure for health effects as absorbable by lungs and bloodstream
- $EV$ and $hybrid$ are number of registered vehicles (/1000) by fuel type in location $\ell$.

## Example: PM2.5

The following estimates use local authority level data for 2019 in the UK.

```
# Load the Stata file
pm25_data <- read.csv(file.path(data_dir, "pm25.csv"))

# Estimate univariate linear regression
model1 <- lm(ln_pm25 ~ ev_k + hybrid_k + petrol_k + diesel_k + ln_pop, data = pm25_data)

# View results
summary(model1)
```

```
Call:
lm(formula = ln_pm25 ~ ev_k + hybrid_k + petrol_k + diesel_k +
    ln_pop, data = pm25_data)

Residuals:
     Min       1Q   Median       3Q      Max
-0.39959 -0.05714  0.03222  0.08413  0.47799

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.265e+00  2.326e-01   5.438 1.16e-07 ***
ev_k         7.798e-02  3.734e-02   2.088   0.0377 *
hybrid_k     3.223e-02  6.352e-03   5.073 7.06e-07 ***
petrol_k     5.591e-05  7.053e-04   0.079   0.9369
diesel_k    -3.546e-03  5.762e-04  -6.154 2.55e-09 ***
ln_pop       8.814e-02  2.084e-02   4.229 3.17e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1304 on 285 degrees of freedom
Multiple R-squared:  0.3698,    Adjusted R-squared:  0.3588
F-statistic: 33.45 on 5 and 285 DF,  p-value: < 2.2e-16
```

## Null hypotheses

From a public policy perspective (if you consider designing subsidies), you want to estimate both $\beta_1$ and $\beta_2$, BUT also know if

$$H_0 : \beta_1 = \beta_2$$

That is, should the policy discriminate bewteen electrified fuel types?

- Engineers may be able provide mechanical differences, but they can't tell you about differences in usage (driver behaviour) which is important for realized pollution mitigation.

> **Warning**
>
> Do not take these estimates too seriously. This model is a classic example of selection into "treatment": those who live in higher pollution areas (e.g. dense urban centres) are also more likely to buy EVs. A better research design would use panel data to evaluate this question dyanmically.

### Alternative hypothesis

The alternative in this case, may be

$$H_1 : \beta_1 \neq \beta_2$$

But may also be,

$$H_1 : \beta_1 < \beta_2$$

or

$$H_1 : \beta_1 > \beta_2$$

if there is a strong case for ruling out one case (see Wooldridge 2025, 139).

### Test statistic

For $H_0 : \beta_1 = \beta_2 \Rightarrow H_0 : \beta_1 - \beta_2 = 0$

$$\text{test-stat} = \frac{\hat{\beta}_1 - \hat{\beta}_2}{\sqrt{Var(\hat{\beta}_1 - \hat{\beta}_2)}}$$

> **Note**
>
> The more general case would be $H_0 : a_1\beta_1 + a_2\beta_2 = \kappa_0$
>
> $$\text{test-stat} = \frac{a_1\hat{\beta}_1 + a_2\hat{\beta}_2 - \kappa_0}{\sqrt{Var(a_1\hat{\beta}_1 + a_2\hat{\beta}_2)}}$$
>
> which can be extended to include more parameters.

## Variance

The key to understanding and executing this test is,[1]

$$Var(\hat{\beta}_1 - \hat{\beta}_2) \neq Var(\hat{\beta}_1) + Var(\hat{\beta}_2)$$

This is because,

$$\begin{aligned} Var(a_1z_1 + a_2z_2) =& Var(a_1z_1) + Var(a_2z_2) + 2 \times Cov(a_1z_1, a_2z_2) \\ =& a_1^2 Var(z_1) + a_2^2 Var(z_2) + 2 \times a_1a_2 Cov(z_1, z_2) \end{aligned}$$

for random variables $z_1$ and $z_2$ and constants $a_1$ and $a_2$.

So,

$$Var(\hat{\beta}_1 - \hat{\beta}_2) = Var(\hat{\beta}_1) + Var(\hat{\beta}_2) - 2\times \underbrace{Cov(\hat{\beta}_1, \hat{\beta}_2)}_{\neq 0}$$

**Why $-2$?**

- For this $H_0$,

$$\Rightarrow H_0 : \underbrace{\beta_1 - \beta_2 = 0}_{a_1\beta_1 + a_2\beta_2 = \kappa_0}$$

  So, $a_1 = 1$ and $a_2 = -1$.

[1] We drop the $Var(\cdot|X)$ notation from Lecture 3. However, all these variances are conditional on the sample values of the independent variables.

### Distribution

Under MLR 1-6, the **conditional** distribution of this test-statistic is $t_{n-k-1}$

$$t\text{-stat} = \frac{\hat{\beta}_1 - \hat{\beta}_2}{\sqrt{Var(\hat{\beta}_1 - \hat{\beta}_2)}} \underset{H_0}{\sim} t_{n-k-1}$$

**under the null**.

- You can use the same steps from Lecture 3 to conduct inference: $p$-value, critical value.
- We tend not to construct confidence intervals in such cases.
- Similarly, you can compute the power under the alternative $\beta_1 - \beta_2 = \kappa$.

### Example: PM2.5

In R, you can use the `marginaleffects` package. However, this is NOT available in AppsAnywhere. Do not worry, there is an alternative coming up.

```
# Load package
library(marginaleffects)

# Estimate linear hypothesis
hypotheses(model1, "ev_k - hybrid_k = 0")
```

```
      Hypothesis Estimate Std. Error    z Pr(>|z|)   S   2.5 % 97.5 %
 ev_k-hybrid_k=0   0.0458     0.0423 1.08     0.28 1.8 -0.0373  0.129
```

### Multiple linear hypotheses

Another common form of hypothesis concerns **multiple** linear combinations of estimators.

$$\begin{aligned} H_0 : a_0^1\beta_0 + a_1^1\beta_1 + \cdots + a_k^1\beta_k &= \kappa_0^1 \\ a_0^2\beta_0 + a_1^2\beta_1 + \cdots + a_k^2\beta_k &= \kappa_0^2 \\ &\vdots \\ a_0^q\beta_0 + a_1^q\beta_1 + \cdots + a_k^q\beta_k &= \kappa_0^q \end{aligned}$$

Linear transformation

You can think of this as a linear transformation of $\beta$.

$$H_0 : R\beta = r$$

Where $R$ is a $q \times k$ matrix of constants and $r$ a $q$-dimensional column vector of constants.

### Example: PM2.5

In our example of

$$PM2.5_\ell = \beta_0 + \beta_1 EV_\ell + \beta_2 hybrid_\ell + \mathbf{x}_\ell\gamma + u_\ell$$

This could be as simple as,

$$H_0 : \beta_1 = 0 \text{ and } \beta_2 = 0$$

Multiple hypothesis testing

Why not test $H_0^{(1)} : \beta_1 = 0$ and then $H_0^{(2)} : \beta_2 = 0$ as separate individual hypotheses? The combined size of these tests would not be $\alpha$.

$$Pr(\text{Reject } H_0^{(1)} \text{ or } H_0^{(2)} | \beta_1 = 0, \beta_2 = 0) \neq \alpha$$

Why? For two reasons:

- Suppose, the tests are independent. Then,

$$\begin{aligned} Pr(\text{Reject } H_0^{(1)} \text{ or } H_0^{(2)} | \beta_1 = 0, \beta_2 = 0) =& Pr(\text{Reject } H_0^{(1)} | \beta_1 = 0, \beta_2 = 0) \\ &+ Pr(\text{Reject } H_0^{(2)} | \beta_1 = 0, \beta_2 = 0) \\ =& 2\alpha \end{aligned}$$

  We could then adjust the size of each individual test to $\alpha/2$ to ensure that the probability of making *at least one* type-1 error is $\alpha$ overall.

- The bigger ("messier") problem is that the tests are not independent of one another because the $Cov(\hat{\beta}_1, \hat{\beta}_2) \neq 0$. This makes it difficult to define this probability.

Multiple hyopthesis testing

There are a number of ways to correctly do multiple hypothesis testing. One is the Bonferroni correction which gets you to reject separate hypotheses sequentially using an algorithim. It does so by ensuring that the family-wise error rate (the probability of rejecting at least one true $H_0$ across a "family" of hypotheses) is $\leq \alpha$.

## Test

The test is performed by considering two models:

1. **Unrestricted** model; using our example,

$$\ln(PM2.5_\ell) = \beta_0 + \beta_1 EV_\ell + \beta_2 hybrid_\ell + \mathbf{x}_\ell\gamma + u_\ell$$

- storing the $SSR_{ur}$ from this model

2. **Restricted model**, which imposes $H_0 : \beta_1 = 0$ and $\beta_2 = 0$,

$$\ln(PM2.5_\ell) = \beta_0 + \mathbf{x}_\ell\gamma + u_\ell$$

- storing the $SSR_r$ from this model

## Test statistic

The test static that is then given by,

$$F\text{-stat} = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/df_{ur}}$$

where,

- $SSR_r$ and $SSR_{ur}$ come from the restricted ($r$) and unrestricted models ($ur$).[2]

$$SSR_r - SSR_{ur} > 0$$

[2]This inequality is how you can remember the order. The $F$-stat is strictly positive.

### Test statistic

The test static that is then given by,

$$F\text{-stat} = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/df_{ur}}$$

and,

- $q$ is the number of restrictions (i.e. equations tested) and is given by the difference in residual degrees of freedom

$$q = df_r - df_{ur}$$

### Distribution

Under the null, $F$-stat has an $F$ distribution (hence the name)

$$F\text{-stat} \underset{H_0}{\sim} F_{q,n-k-1}$$

We reject $H_0$ if

$$F\text{-stat} > F^{-1}_{q,n-k-1}(1-\alpha) = f_{q,n-k-1,1-\alpha}$$

where $F^{-1}_{q,n-k-1}(1-\alpha)$ is the $1-\alpha$ percentile of the distribution.

### Rejection rule

$$F\text{-stat} > f_{q,n-k-1,1-\alpha}$$

- **Always a one-sided test**.[3]
- Doesn't mean that $H_1$ is one-sided.
  - You cannot tell which direction the deviation from the null is, because F-stat uses squared-deviations (i.e. sign is lossed) .
- $H_1:$ $H_0$ false, or $H_1 : \beta_1 \neq 0$ or $\beta_2 \neq 0$

[3]This does not mean that $F$-distributed tests are always one-sided, but rather that multiple linear hypotheses of this form are.

*F* distribution

The $F$ distribution is defined as the ratio of two chi-squared distributions,

$$F_{q,m} = \frac{\chi^2_q/q}{\chi^2_m/m}$$

each normalized by their corresponding degrees of freedom.

### Alternative form

Since $SSR = (1 - \mathbf{R}^2)SST$ and $SST_r = SST_{ur}$,

We can write the $F$-stat as,

$$F\text{-stat} = \frac{(\mathbf{R}^2_{ur} - \mathbf{R}^2_r)/q}{(1 - \mathbf{R}^2_{ur})/(n - k - 1)}$$

Note,

- The order switches in the numerator, but remains positive since $\mathbf{R}^2_{ur} > \mathbf{R}^2_r$
- $n - k - 1 = df_{ur}$, with $k$ the number of parameters in the unrestricted model

### Example: PM2.5

In R, you can use the `car` package, which is available in AppsAnywhere.

```
# Load package
library(car)

# Estimate linear hypothesis
linearHypothesis(model1, c("ev_k = 0", "hybrid_k = 0"))
```

```
Linear hypothesis test:
ev_k = 0
hybrid_k = 0

Model 1: restricted model
Model 2: ln_pm25 ~ ev_k + hybrid_k + petrol_k + diesel_k + ln_pop

  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
```

```
1    287 6.6816
2    285 4.8484  2    1.8332 53.881 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

### Revisiting single linear restriction

You can think of a single linear hypothesis as a special case of multiple where $q = 1$. For example,

$$H_0 : \beta_1 = \beta_2$$

- Restricted model:

$$\begin{aligned}\ln(PM2.5_\ell) =& \beta_0 + \beta_1 EV_\ell + \underbrace{\beta_1}_{=\beta_2} hybrid_\ell + \mathbf{x}_\ell\gamma + u_\ell \\ =& \beta_0 + \beta_1(EV_\ell + hybrid_\ell) + \mathbf{x}_\ell\gamma + u_\ell\end{aligned}$$

  A regression with a new transformed variable: $EV_\ell + hybrid_\ell$

Moreover, you can show that for $q = 1$

$$F\text{-stat} = \frac{SSR_r - SSR_{ur}}{SSR_{ur}/(n-k-1)} = (t\text{-stat})^2$$

### Example: PM2.5

```r
# t-test
hypotheses(model1, "ev_k - hybrid_k = 0")
```

```
      Hypothesis Estimate Std. Error    z Pr(>|z|)   S   2.5 % 97.5 %
 ev_k-hybrid_k=0   0.0458     0.0423 1.08     0.28 1.8 -0.0373  0.129
```

```r
# F-test
linearHypothesis(model1, "ev_k = hybrid_k")
```

```
Linear hypothesis test:
ev_k - hybrid_k = 0

Model 1: restricted model
Model 2: ln_pm25 ~ ev_k + hybrid_k + petrol_k + diesel_k + ln_pop

  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    286 4.8682
2    285 4.8484  1  0.019855 1.1671 0.2809
```

## Regression significance

A special case of this form of test concerns the joint significance of all the *slope* parameters in a model:

$$H_0 : \beta_1 = 0,\ \beta_2 = 0,\ \ldots,\ \beta_k = 0$$

In this case,

- The restricted model is a regression on a constant (alone) where $\Rightarrow \mathbf{R}^2_r = 0$
- And $\mathbf{R}^2_{ur}$ is simply the $\mathbf{R}^2$ from the model
- The number of restrictions is $q = k$

We define the $F$-stat as,

$$F\text{-stat} = \frac{\mathbf{R}^2/k}{(1-\mathbf{R}^2)/(n-k-1)}$$

## Stata output

This test is reported as standard with the Stata output as:

- `F( , )`: $F$-stat
- `Prob > F`: corresponding $p$-value of $F$-test

```
no variables defined
r(111);

r(111);
```

## Structural Change

### Dummy variables: recap

Recall, when we have the model:

$$y_i = \beta_0 + \beta_1 d_i + u_i$$

where $d_i \in \{0, 1\}$. Then,

$$\beta_1 = E[y_i|d_i = 1] - E[y_i|d_i = 0]$$

### Dummy variable interactions

Now, suppose the model was

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 d_i + \beta_3 x_i \times d_i + u_i$$

where $x_i$ is continuous.

- What is the interpretation of each parameter?

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 d_i + \beta_3 x_i \times d_i + u_i$$

- $\beta_0 = E[y_i|d_i = 0, x_i = 0]$
- $\beta_1 = \frac{dE[y_i|d_i=0,x_i]}{dx_i}$
- $\beta_2 = E[y_i|d_i = 1, x_i = 0] - E[y_i|d_i = 0, x_i = 0]$
- $\beta_3 = \frac{dE[y_i|d_i=1,x_i]}{dx_i} - \frac{dE[y_i|d_i=0,x_i]}{dx_i}$

Another way of thinking about it:

- For $d_i = 0$, the model is:

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

- For $d_i = 1$, the model is:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 + \beta_3 x_i + u_i$$

  Which can be rewritten as:

$$y_i = \underbrace{(\beta_0 + \beta_2)}_{\gamma_0} + \underbrace{(\beta_1 + \beta_3)}_{\gamma_1} x_i + u_i$$

We have two models: for $d_i = 0$ and $d_i = 1$.

- $\beta_2$ is the difference in intercept
- $\beta_3$ is the difference in slope

### Example

The typical example is the returns to education for men and women (as in Wooldridge 2025, 243)

- But, given what we know about the 'parent penalty' (Kleven et al. 2019), it seems more pertinent to focus on experience

$$\ln(wage_i) = \beta_0 + \beta_1 exp_i + u_i$$

```
wage_data <- read_dta(file.path(data_dir, "morg24_bothgen.dta"))

ggplot(wage_data, aes(x = exp, y = lnwage)) +
  geom_point(colour = "blue", alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 1.5) +
  facet_wrap(~ as_factor(sex)) +
  labs(x = "Potential Experience (=Age-(Years of Education +6))") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



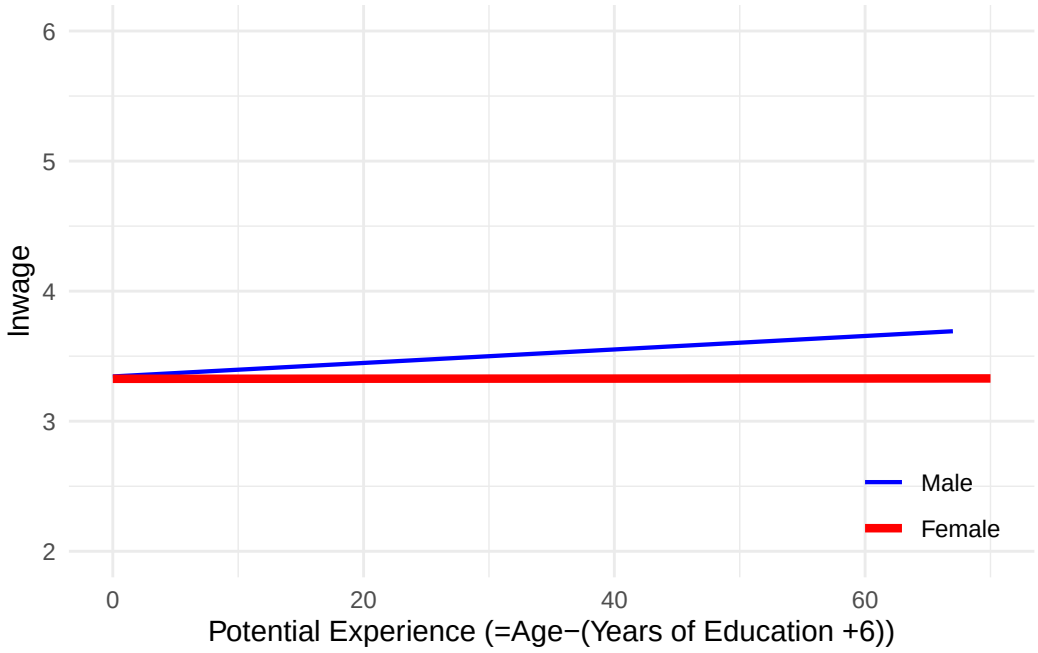

If we overlay the fitted lines we get

```
wage_data$group <- factor(wage_data$sex, levels = c(1, 2),
                          labels = c("Male", "Female"))

ggplot(wage_data, aes(x = exp, y = lnwage, colour = group)) +
  geom_smooth(data = subset(wage_data, sex == 1),
              method = "lm", se = FALSE, linewidth = 0.75) +
  geom_smooth(data = subset(wage_data, sex == 2),
```

```
              method = "lm", se = FALSE, linewidth = 1.5) +
  scale_colour_manual(values = c(Male = "blue", Female = "red")) +
  scale_y_continuous(breaks = 2:6) +
  coord_cartesian(ylim = c(2, 6)) +
  labs(x = "Potential Experience (=Age-(Years of Education +6))",
       y = "lnwage", colour = NULL) +
  theme_minimal() +
  theme(legend.position = "inside",
        legend.position.inside = c(0.98, 0.02),
        legend.justification = c(1, 0))
```

```
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```




### $F$-test

The $F$-test can be used to test differences in a model across groups.[4]

- For example, whether the Mincer equation (from Tutorial 1) differs by gender.

[4]We will look at the case of two groups, but the approach can be extended to multiple groups.

$$\ln(wage_i) = \beta_0 + \beta_1 edu_i + \beta_2 exp_i + u_i$$

- We estimated this model for the male-only sample
- The quadratic term in *exp* has been dropped for simplification
- The model has $k + 1 = 3$ parameters

### Test:

We may wish to ask whether:

- the model for men (group = 1; e.g. "Male")

$$\ln(wage_i) = \beta_0 + \beta_1 edu_i + \beta_2 exp_i + u_i$$

is the same as,

- the model for women (group = 2; e.g. "Female")?

$$\ln(wage_i) = \gamma_0 + \gamma_1 edu_i + \gamma_2 exp_i + u_i$$

### Null hypothesis

This question implies a null hypothesis:

$$H_0 : \beta_0 = \gamma_0,\ \beta_1 = \gamma_1,\ \beta_2 = \gamma_2$$

against,

$$H_1 : H_0 \text{ false}$$

## Chow test

The relevant $F$-statistic for this test is referred to as the Chow statistic:

$$F\text{-stat} = \frac{(SSR_P - (SSR_1 + SSR_2))/(k+1)}{(SSR_1 + SSR_2)/(n-2(k+1))}$$

where,

- $SSR_P$ is the "pooled" ($P$) $SSR$ from a regression of the joint sample.
- $SSR_1$ is the $SSR$ from a regression for just group 1.
- $SSR_2$ is the $SSR$ from a regression for just group 2.
  - together $SSR_1$ and $SSR_2$ represent the unrestricted model.
- $n - 2(k+1)$ is the **combined** degrees of freedom from group 1 and 2.
  - $k+1$ parameters are estimated in each group.

## Distribution

The $F$-stat is (unsurprisingly) $F$-distributed,

$$F\text{-stat} \underset{H_0}{\sim} F_{k+1, n-2(k+1)}$$

## Approach 2

This test has an equivalent form using interactions. Let,

$$d_i = \mathbf{1}\{\text{'Group 2'}\}$$

For example,

$$d_i = \mathbf{1}\{\text{'Female'}\}$$

## Interactions

We can write the unrestricted model as,

$$\ln(wage_i) = \beta_0 + \beta_1 edu_i + \beta_2 exp_i + \beta_3 d_i + \beta_4 edu_i \times d_i + \beta_5 exp_i \times d_i + u_i$$

We now have 6 parameters: $2 \times (k+1)$ (for $k = 2$)

As before, the dummy and dummy-interactions tell you about differences,

$$\beta_3 = E[\ln(wage_i)|d_i = 1, edu_i = 0, exp_i = 0] - E[\ln(wage_i)|d_i = 0, edu_i = 0, exp_i = 0]$$

and

$$\beta_j = \frac{\partial E[\ln(wage_i)|d_i = 1, edu_i, exp_i]}{\partial x_{ij}} - \frac{\partial E[\ln(wage_i)|d_i = 0, edu_i, exp_i]}{\partial x_{ij}} \qquad j = 4, 5$$

## Test

Intuitively, the test for whether the model differs by group is given by,

$$H_0 : \beta_3 = 0,\ \beta_4 = 0,\ \beta_5 = 0$$

where,

- **unrestricted** model is given by,

$$\ln(wage_i) = \beta_0 + \beta_1 edu_i + \beta_2 exp_i + \beta_3 d_i + \beta_4 edu_i \times d_i + \beta_5 exp_i \times d_i + u_i$$

- $df_{ur} = n - 6$

- **restricted** model is given by,

$$\ln(wage_i) = \beta_0 + \beta_1 edu_i + \beta_2 exp_i + v_i$$

- $df_r = n - 3$

### $F$-stat

The $F$-stat is given by,

$$F\text{-stat} = \frac{(SSR_r - SSR_{ur})/3}{SSR_{ur}/(n-6)}$$

Compare this to the first approach:

$$F\text{-stat} = \frac{(SSR_P - (SSR_1 + SSR_2))/3}{(SSR_1 + SSR_2)/(n-6)}$$

### Equivalence

These two test-statics are equivalent because,

1. $SSR_P = SSR_r$
2. $SSR_1 + SSR_2 = SSR_{ur}$

### Demonstration

Here are the SSR for each version:

```
# make sex a factor variable
wage_data$sex <- factor(wage_data$sex)
```

- Model for group 1 (i.e. "male")

```
model_m <- lm(lnwage ~ edu + exp, data = wage_data, subset = sex == 1)
deviance(model_m)
```

```
[1] 715.9086
```

- Model for group 2 (i.e. "female")

```
model_f <- lm(lnwage ~ edu + exp, data = wage_data, subset = sex == 2)
deviance(model_f)
```

```
[1] 539.5977
```

- Unrestricted model (i.e. with interactions)

```
model_ur <- lm(lnwage ~ sex * edu + sex * exp, data = wage_data)
deviance(model_ur)
```

```
[1] 1255.506
```

- Restricted model (i.e. without interactions)

```
model_r <- lm(lnwage ~ edu + exp, data = wage_data)
deviance(model_r)
```

```
[1] 1307.167
```

Here is how you would programme the test

- Estimate interacted model

```
model_ur <- lm(lnwage ~ sex * edu + sex * exp, data = wage_data)
linearHypothesis(model_ur, c("sex2 = 0", "sex2:edu = 0", "sex2:exp = 0"))
```

```
Linear hypothesis test:
sex2 = 0
sex2:edu = 0
sex2:exp = 0

Model 1: restricted model
Model 2: lnwage ~ sex * edu + sex * exp
```

```
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1   4997 1307.2
2   4994 1255.5  3     51.66 68.496 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The notation `sex2` is how R names the coefficient on level `2` of the factor `sex`, and `sex2:edu` its interaction with `edu` (a continuous variable).

# References

## Bibliography


Kleven, Henrik, Camille Landais, Johanna Posch, Andreas Steinhauer, and Josef Zweimüller. 2019. "Child Penalties Across Countries: Evidence and Explanations." In *Aea Papers and Proceedings*, 109:122–26. American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203.

Wooldridge, Jeffrey M. 2025. "Introductory Econometrics: A Modern Approach."